

Monthly crime trends: a global STL-style decomposition

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What this project estimates

This is a descriptive decomposition of monthly crime counts for every city in the RTCI national sample. Rates below are annualized from monthly counts, so a monthly murder count of (m) in a population (N) is shown as $(12m/N \times 100,000)$. This is why the homicide values are on the familiar annual rate scale rather than below one.

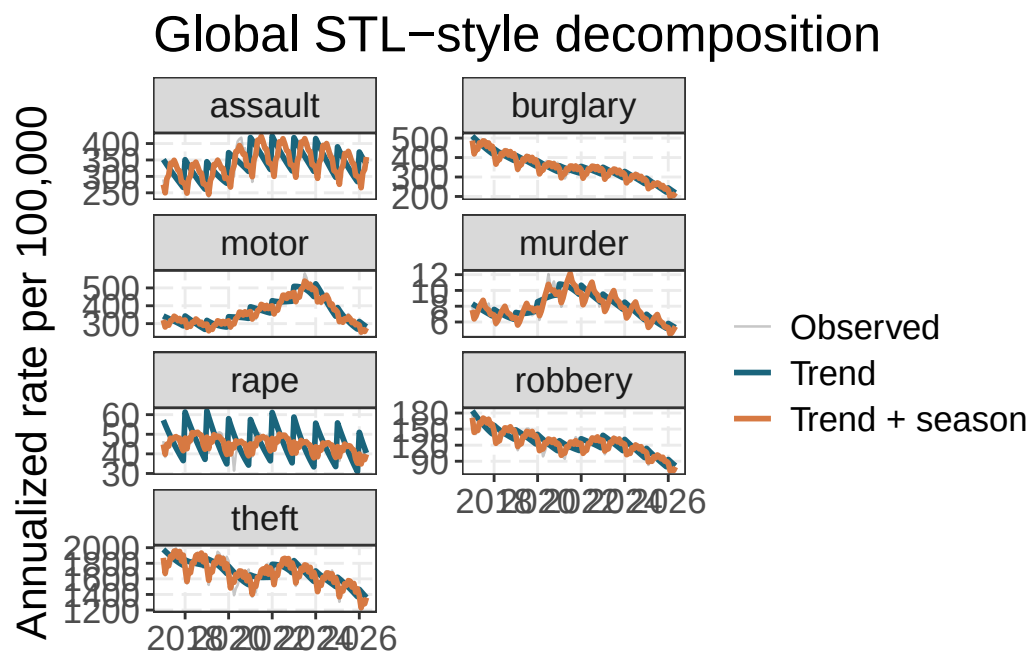
The model uses a single stacked aggregated-binomial response for murder, rape, robbery, assault, burglary, theft, and motor-vehicle theft. It estimates a crime-specific smooth long-run trend, a smooth year term, and a cyclic month-of-year seasonal term. City and city-by-crime effects can be enabled for focused runs. The national deliverable uses the stable global fit for every city, then reports the requested city $\times$ crime $\times$ month overdispersion term as the logit departure of the observed cell from that baseline.

The source is the pinned Real-Time Crime Index snapshot (AH-Datalytics 2026), and the aggregated-binomial motivation follows Wheeler and Kovandzic (Wheeler and Kovandzic 2018).

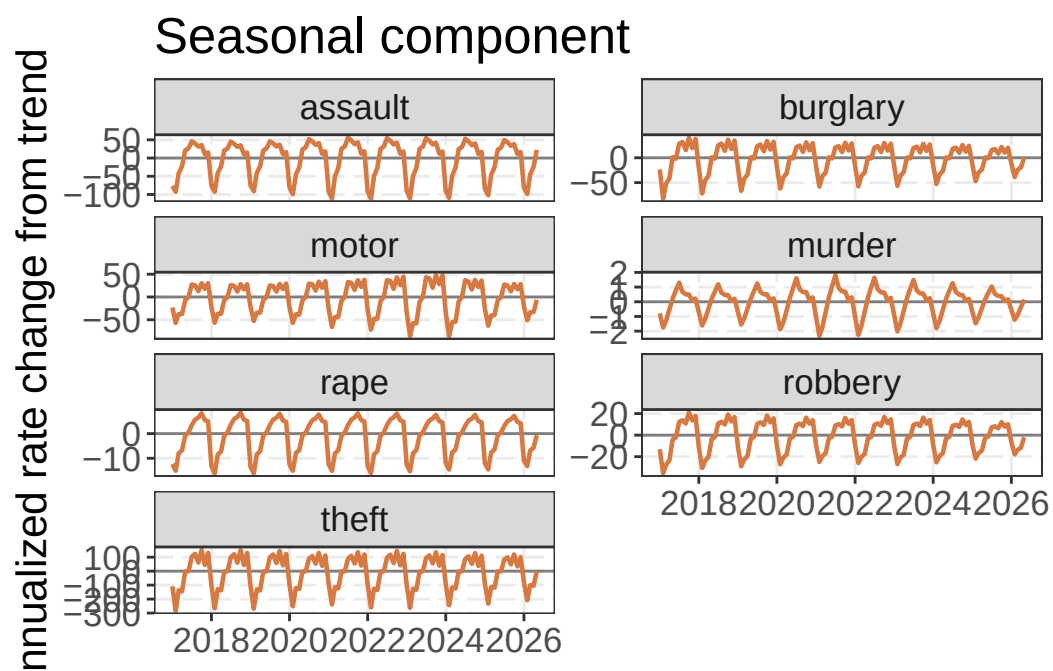
Data coverage

The deliverable contains 458,392 valid city-month-crime observations from 586 cities. It includes no population threshold. The city-name and coordinate crosswalk comes from the RTCI processed agency metadata.

Global STL-style decomposition

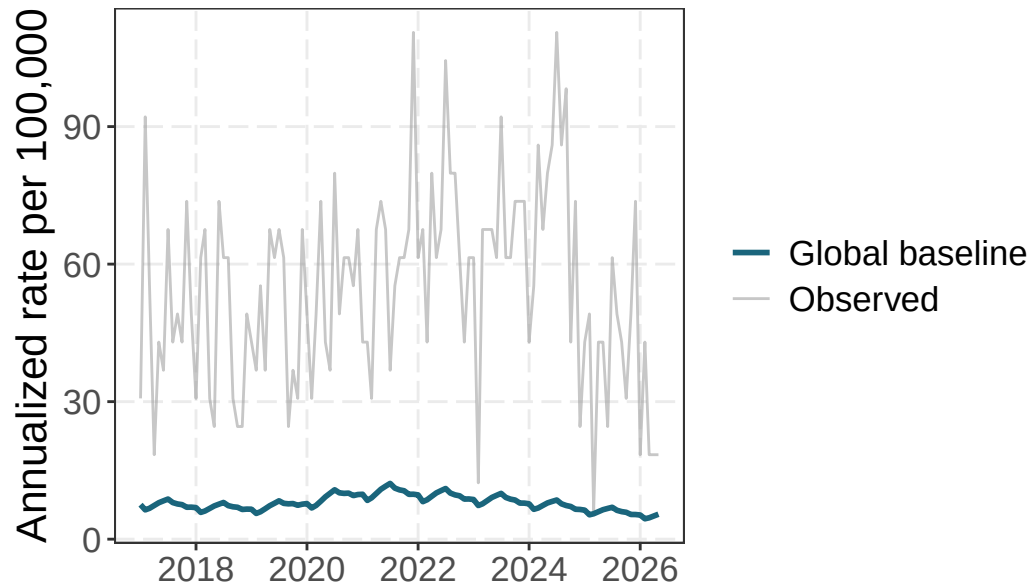


The legend is explicit: grey is the observed population-weighted rate, blue is the long-run smooth trend, and orange is trend plus the cyclic seasonal effect.

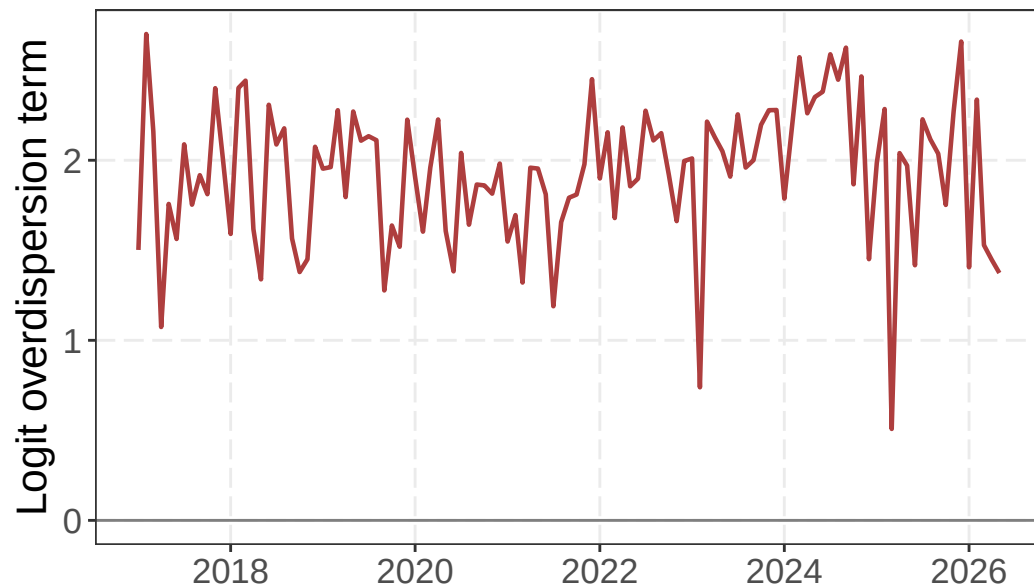


City detail and overdispersion

Birmingham, AL — murder



City × crime × month overdispersion



The overdispersion term is not a Pearson residual and is not filtered by an arbitrary threshold. It is retained at the city, crime, and month level so the app can show the complete time series and let the analyst decide which departures matter.

Interpretation and next steps

The decomposition is descriptive, not causal. Reporting completeness, population estimates, agency coverage, and revisions can all affect the curves. The next useful extensions are uncertainty intervals for the global

and city terms, a formal shrinkage model for the city $\times$ crime $\times$ month effect, and a sensitivity analysis for reporting changes.

AH-Datalytics. 2026. “Real-Time Crime Index Repository.” <https://github.com/AH-Datalytics/rtci>.

Wheeler, Andrew P., and Tomislav V. Kovandzic. 2018. “Monitoring Volatile Homicide Trends Across US Cities.” *Homicide Studies* 22 (2): 119–44.